Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Artificial Intelligence (01CT0616)	**Aim:** To study Preprocessing of text (Tokenization. Filtration, Script Validation, Stop Word Removal, Stemming)	
Experiment No: 6	**Date: 18/03/2026**	**Enrolment No: 92410133038**

Aim: To study Preprocessing of text (Tokenization. Filtration, Script Validation, Stop Word Removal, Stemming)

IDE: Google Colab

Theory:

Text preprocessing is the first and most important step in Natural Language Processing (NLP). It means converting raw text into a clean and understandable format so that machines can process it easily.
Since machine learning models cannot directly understand text data, we first need to convert text into numerical form using techniques like Bag of Words, n-grams, TF-IDF, or Word2Vec. But before that, the text must be cleaned properly. This cleaning process is called text preprocessing.
Different preprocessing steps help in improving the quality of data and removing unnecessary information from the text.

Program (Code):

```
text="""I am Aravind Miriyampalli. I pursuing B.Tech in ICT-Sem 6. I interested

in VLSI domain and I find my project in A.I and HCD domain."""


#spliting in the words

text.split()


##for NLP there is 3 library 'spacy, gensim, nltk'

import spacy

nlp=spacy.load('en_core_web_sm')


doc=nlp(text)
```

```
doc[5]

# NER: Named Entity Recognition

doc = nlp("Your sentence here")

for tiken in doc.ents:

  print(token.text, token.label_)



#Lemmatization vs stemming

text="""I studied the subject AI and then

meeting My Banneshu in the meeting"""



import nltk

from nltk.stem.porter import *  ## stemming

p_stem=PorterStemmer()

for word in text.split():

  print(word,"-->", p_stem.stem(word))


#lemmatization

for token in nlp(text):

  print(token.text,"- ",token.pos_,
```

```
        "- ",token.lemma_)


#Lemmatization vs stemming

text="""I studied the subject AI and then

meeting My Banneshu in the meeting"""



import nltk

from nltk.stem.porter import *  ## stemming

p_stem=PorterStemmer()

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for token in nlp(text):

  print(token.text,"- ",token.pos_,

        "- ",token.lemma_)
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Results:

```
[31]
text="""I am Manasa Rathod. I pursuing B.Tech in ICT-Sem 6. I interested
in VLSI domain and I find my project in A.I and HCD domain."""

[32]
#spliting in the words
text.split()

['I',
 'am',
 'Manasa',
 'Rathod.',
 'I',
 'pursuing',
 'B.Tech',
 'in',
 'ICT-Sem',
 '6.',
 'I',
 'interested',
 'in',
 'VLSI',
 'domain',
 'and',
 'I',
 'find',
 'my',
 'project',
 'in',
 'A.I',
 'and',
 'HCD',
```

```
[19]
##for NLP there is 3 library 'spacy, gensim, nltk'
import spacy
nlp=spacy.load('en_core_web_sm')

[21]
doc=nlp(text)

doc[5]
# NER: Named Entity Recognition
doc = nlp("Your sentence here")
for tiken in doc.ents:
  print(token.text, token.label_)

[22]
#Lemmatization vs stemming
text="""I studied the subject AI and then
meeting My Banneshu in the meeting"""

[23]
import nltk
from nltk.stem.porter import *  ## stemming
p_stem=PorterStemmer()
for word in text.split():
  print(word,"-->", p_stem.stem(word))
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```
I --> i
studied --> studi
the --> the
subject --> subject
AI --> ai
and --> and
then --> then
meeting --> meet
My --> my
Banneshu --> banneshu
in --> in
the --> the
meeting --> meet
```

```python
#lemmatization
for token in nlp(text):
  print(token.text,"- ",token.pos_,
        "- ",token.lemma_)
```

```
I -  PRON -  I
studied -  VERB -  study
the -  DET -  the
subject -  ADJ -  subject
AI -  PROPN -  AI
and -  CCONJ -  and
then -  ADV -  then

 -  SPACE -

meeting -  VERB -  meet
My -  PRON -  my
```

Terminal 10:08 AM Python 3

```
meeting -  VERB -  meet
My -  PRON -  my
Banneshu -  NOUN -  banneshu
in -  ADP -  in
the -  DET -  the
meeting -  NOUN -  meeting
```

[26] 0s

```python
#Lemmatization vs stemming
text="""I studied the subject AI and then
meeting My Banneshu in the meeting"""
```

[27] 0s

```python
import nltk
from nltk.stem.porter import *  ## stemming
p_stem=PorterStemmer()
for word in text.split():
  print(word,"-->", p_stem.stem(word))
```

```
I --> i
studied --> studi
the --> the
subject --> subject
AI --> ai
and --> and
then --> then
meeting --> meet
My --> my
```

Terminal 10:08 AM Python 3

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```
then --> then
meeting --> meet
My --> my
Banneshu --> banneshu
in --> in
the --> the
meeting --> meet
```

[28] 0s

```
#lemmatization
for token in nlp(text):
  print(token.text,"- ",token.pos_,
        "- ",token.lemma_)
```

```
I -  PRON -  I
studied -  VERB -  study
the -  DET -  the
subject -  ADJ -  subject
AI -  PROPN -  AI
and -  CCONJ -  and
then -  ADV -  then

 -  SPACE -

meeting -  VERB -  meet
My -  PRON -  my
Banneshu -  NOUN -  banneshu
in -  ADP -  in
the -  DET -  the
meeting -  NOUN -  meeting
```

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Observation:

After performing the experiment, I observed that text preprocessing helps in cleaning and simplifying the raw text data. Tokenization divides the text into smaller parts, making it easier to analyze.
Stop word removal reduces unnecessary words and improves efficiency. Stemming helps in grouping similar words by converting them into their root form, although sometimes the output is not a proper word.
Overall, preprocessing improves the quality of data and makes it suitable for further NLP tasks.

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Post Lab Exercise:

1. **Take your own document of 10 sentences and perform the same tasks. Attach code and screenshot of your output.**

Program(Code):-

```
text = """I am Manasa Rathod. I am pursuing B.Tech in ICT Semester 6.
I am interested in VLSI domain. I like working on Artificial Intelligence.
My project is based on AI and HCD. I enjoy learning new technologies eventually.
I often practice coding in Python and exploring on it. My goal is to become a good engineer and being independent.
I also like teamwork and collaboration. I am improving my problem solving skills and my innovation skills."""

print("Word Splitting:\n", text.split())
import spacy
nlp = spacy.load('en_core_web_sm')

doc = nlp(text)
print("\n5th Token:", doc[5])

# NER: Named Entity Recognition
print("\nNamed Entities:")
for token in doc.ents:
    print(token.text, "-->", token.label_)

# LEMMATIZATION vs STEMMING
text2 = """I studied the subject AI and then
meeting my friend in the meeting"""

# STEMMING (NLTK)
import nltk
from nltk.stem.porter import PorterStemmer
p_stem = PorterStemmer()

print("\nStemming Output:")
for word in text2.split():
    print(word, "-->", p_stem.stem(word))

# LEMMATIZATION (spaCy)
print("\nLemmatization Output:")
doc2 = nlp(text2)
for token in doc2:
    print(token.text, "-", token.pos_, "-", token.lemma_)
```

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Screenshots of Codes & Outputs:-

NLP own sentence

```
[2]
text = """I am Manasa Rathod. I am pursuing B.Tech in ICT Semester 6.
I am interested in VLSI domain. I like working on Artificial Intelligence.
My project is based on AI and HCD. I enjoy learning new technologies eventually.
I often practice coding in Python and exploring on it. My goal is to become a good engineer and being independent.
I also like teamwork and collaboration. I am improving my problem solving skills and my innovation skills."""
```

```
[3]
print("Word Splitting:\n", text.split())
```

```
Word Splitting:
 ['I', 'am', 'Manasa', 'Rathod.', 'I', 'am', 'pursuing', 'B.Tech', 'in', 'ICT', 'Semester', '6.', 'I', 'am', 'interested', 'in', 'VLSI', 'domain.', 'I', 'like', 'working', 'o
```

```
[4]
import spacy
nlp = spacy.load('en_core_web_sm')

doc = nlp(text)
```

```
[5]
print("\n5th Token:", doc[5])
```

```
5th Token: I
```

```
[9]
# NER: Named Entity Recognition
print("\nNamed Entities:")
for token in doc.ents:
    print(token.text, "-->", token.label_)
```

NLP own sentence

```
Named Entities:
Aravind Miriyampalli > PERSON
B.Tech --> ORG
ICT Semester --> ORG
6 --> CARDINAL
Artificial Intelligence --> ORG
AI --> GPE
HCD --> ORG
Python --> GPE
```

```
[10]
# LEMMATIZATION vs STEMMING
text2 = """I studied the subject AI and then
meeting my friend in the meeting"""
```

```
[11]
# STEMMING (NLTK)
import nltk
from nltk.stem.porter import PorterStemmer
```

```
[12]
p_stem = PorterStemmer()
```

```
[13]
print("\nStemming Output:")
for word in text2.split():
    print(word, "-->", p_stem.stem(word))
```

```
Stemming Output:
I --> i
```

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NLP own sentence

```
the --> the
subject --> subject
AI --> ai
and --> and
then --> then
meeting --> meet
my --> my
friend --> friend
in --> in
the --> the
meeting --> meet
```

[16]

```
# LEMMATIZATION (spaCy)
print("\nLemmatization Output:")
doc2 = nlp(text2)
for token in doc2:
    print(token.text, "-", token.pos_, "-", token.lemma_)
```

```
Lemmatization Output:
I - PRON - I
studied - VERB - study
the - DET - the
subject - ADJ - subject
AI - PROPN - AI
and - CCONJ - and
then - ADV - then

 - SPACE -

meeting - VERB - meet
```

NLP own sentence

```
meeting --> meet
```

[16]

```
# LEMMATIZATION (spaCy)
print("\nLemmatization Output:")
doc2 = nlp(text2)
for token in doc2:
    print(token.text, "-", token.pos_, "-", token.lemma_)
```

```
Lemmatization Output:
I - PRON - I
studied - VERB - study
the - DET - the
subject - ADJ - subject
AI - PROPN - AI
and - CCONJ - and
then - ADV - then

 - SPACE -

meeting - VERB - meet
my - PRON - my
friend - NOUN - friend
in - ADP - in
the - DET - the
meeting - NOUN - meeting
```

2. **Write your observation for stemming and lemmatization you obtained for the sentences.**

From the experiment, I observed that stemming simply removes the ending of words using basic rules, so sometimes the output is not meaningful (like “troubl” instead of “trouble”).
On the other hand, lemmatization produces proper root words, but it is more complex and requires understanding of the word context.
So, stemming is faster but less accurate, while lemmatization is more accurate but slower.
Results
The preprocessing techniques like tokenization, filtration, stop word removal, POS tagging, noun phrase chunking, and dependency parsing were successfully implemented, and the outputs were obtained.

Conclusion:-

This experiment helped me understand the importance of text preprocessing in NLP. It is a necessary step before applying any machine learning model on text data. Techniques like tokenization, stop word removal, and stemming help in improving data quality and reducing complexity